

Practice Midterm 2 — Answer Key

CSCI 160: Computer Architecture 1

Part I: Multiple Choice. 1. B 2. C 3. B 4. C 5. C 6. A

Part II: Show Work

1. $F(a, b, c) = a'c + ab' + bc = \sum m(1, 3, 4, 5, 7)$.

Truth table (row order $abc = 000 \dots 111$): $F = 0, 1, 0, 1, 1, 1, 0, 1$.

Grouping: the column $c = 1$ (cells m_1, m_3, m_5, m_7) gives c ; the bottom-left pair m_4, m_5 gives ab' .

$$F = c + ab'$$

$a \backslash bc$	00	01	11	10
0	0	1	1	0
1	1	1	1	0

2. $F(w, x, y, z) = \sum m(3, 4, 5, 6, 7, 11, 15)$.

Group m_4, m_5, m_6, m_7 (row $wx = 01$) $\Rightarrow w'x$. Group m_3, m_7, m_{11}, m_{15} (column $yz = 11$) $\Rightarrow yz$.

$$F = w'x + yz$$

$wx \backslash yz$	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	0	0	1	0
10	0	0	1	0

3. $F(a, b, c, d) = \sum m(0, 2, 3, 9, 13)$, $d = (1, 11, 15)$.

Use don't-care 1 with m_0, m_2, m_3 to fill row $ab = 00 \Rightarrow a'b'$. The 1's m_9, m_{13} with don't-cares m_{11}, m_{15} form column $cd = 11$ at $a = 1 \Rightarrow ad$.

$$F = a'b' + ad$$

$ab \backslash cd$	00	01	11	10
00	1	X	1	1
01	0	0	0	0
11	0	1	X	0
10	0	1	X	0

4. BCD, $F(a, b, c, d) = \sum m(0, 2, 4, 6, 8)$, $d = (10, \dots, 15)$.

Every 1 has $d = 0$. Including the don't-cares m_{10}, m_{12}, m_{14} lets the entire $d = 0$ region group together.

$$F = d'$$

$ab \backslash cd$	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	X	X
10	1	0	X	X

5. $F(w, x, y, z) = \sum m(2, 3, 6, 7, 8, 9, 10, 11, 15)$.

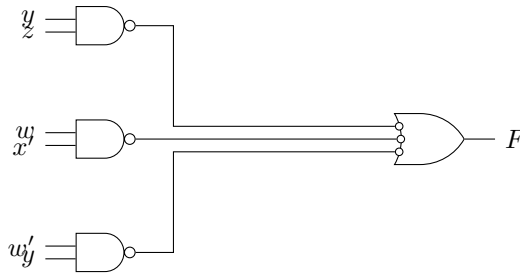
Group m_2, m_3, m_6, m_7 (at $w = 0, y = 1$) $\Rightarrow w'y$. Group m_8, m_9, m_{10}, m_{11} (row $wx = 10$) $\Rightarrow wx'$.

Group m_3, m_7, m_{11}, m_{15} (column $yz = 11$) $\Rightarrow yz$.

$$F = yz + wx' + w'y$$

$wx \backslash yz$	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	0
10	1	1	1	1

NAND form (two faces of NAND: every AND $\rightarrow$ AND-INVERT, every OR $\rightarrow$ INVERT-OR):



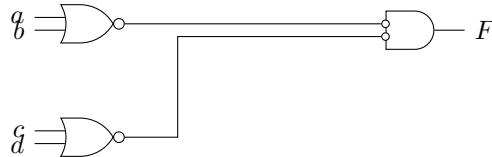
6. $F(a, b, c, d) = \prod M(0, 1, 2, 3, 4, 8, 12)$.

Grouping the 0's (maxterms): m_0, m_1, m_2, m_3 (row $ab = 00$) $\Rightarrow a'b'$; m_0, m_4, m_8, m_{12} (column $cd = 00$) $\Rightarrow c'd'$. So $F' = a'b' + c'd'$, and by DeMorgan

$$F = (a + b)(c + d)$$

$ab \backslash cd$	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	0	1	1	1
10	0	1	1	1

NOR form (two faces of NOR: every OR $\rightarrow$ OR-INVERT, every AND $\rightarrow$ INVERT-AND):



7.

(a) $a \oplus 0 \oplus a \oplus b$:

$$a \oplus 0 \oplus a \oplus b = (a \oplus a) \oplus 0 \oplus b = 0 \oplus 0 \oplus b = b.$$

(Used $a \oplus a = 0$ and $0 \oplus b = b$.) b

(b) $A \oplus B \oplus C = \sum m(1, 2, 4, 7)$.

Reason: a chain of XORs equals 1 exactly when an *odd* number of inputs are 1. The minterms with an odd count of 1's are 001, 010, 100, 111, i.e. 1, 2, 4, 7.

A	B	C	# of 1's	$A \oplus B \oplus C$
0	0	0	0 (even)	0
0	0	1	1 (odd)	1
0	1	0	1 (odd)	1
0	1	1	2 (even)	0
1	0	0	1 (odd)	1
1	0	1	2 (even)	0
1	1	0	2 (even)	0
1	1	1	3 (odd)	1

(c) $\text{Equal} = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$.

Each XNOR ($\odot$) outputs 1 only when its two bits match; the AND is 1 only when every position matches.

For $A = 1011$, $B = 1001$: bits agree at positions 3, 2, 0 but differ at position 1 ($A_1 = 1$, $B_1 = 0$), so $(A_1 \odot B_1) = 0$ and $\text{Equal} = 0$ — the numbers are *not* equal.

(d) Message $wxyz = 1101$ has three 1's (odd). Even parity requires an even total, so $P = 1$:

$$P = w \oplus x \oplus y \oplus z = 1, \quad wxyzP = \boxed{11011}.$$